ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MATHEMATICAL LITERACY

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 4. Finance – home loans, investments, inflation, business statements and income tax (35), measurement (20), maps and route planning (15), data handling (20), probability and risk (10). All people, prices and data are fictional and were created for this examination.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations. Answers only will not necessarily earn full marks.
4. An approved calculator (non-programmable and non-graphical) may be used.
5. Round off answers to TWO decimal places, unless stated otherwise. Write units with all answers.
6. Write neatly and legibly.

QUESTION 1: FINANCE

1.1 Mr and Mrs Khumalo want to take out a home loan of R850 000 at 11,75% per annum. The bank gives them this table of monthly repayments for every R1 000 borrowed:

Loan term	Monthly repayment per R1 000
20 years	R10,84
30 years	R10,09

1.1.1 Calculate the monthly repayment for each loan term. (3)

1.1.2 Calculate the total amount repaid for each loan term. (3)

1.1.3 How much more will they pay in total over 30 years than over 20 years? Advise them which term to choose. (2)

1.2 Ayanda has R20 000 to invest for 5 years. Option 1: 7,5% per annum compounded monthly. Option 2: 8% per annum simple interest.

1.2.1 Calculate the final amount for each option. (4)

1.2.2 Which option is better, and by how much? (2)

1.3 Zinhle earns R15 000 per month. She receives a salary increase of 4%, while inflation for the year is 6%.

1.3.1 Calculate her new monthly salary. (2)

1.3.2 Goods that cost R15 000 last year cost how much now? (2)

1.3.3 Explain whether Zinhle is better off or worse off after the increase. (2)

1.4 A small catering business recorded the following for March: sales R48 500; cost of sales R21 300; rent R6 000; salaries R12 000; other expenses R2 450.

1.4.1 Calculate the gross profit. (2)

1.4.2 Calculate the net profit. (3)

1.4.3 Calculate the net profit as a percentage of sales. (3)

1.5 Use the simplified tax table below to calculate the monthly income tax of a person under 65 whose taxable income is R420 000 per year.

Taxable income (R per year)	Rate of tax

237 101 – 370 500	42 678 + 26% of taxable income above 237 100
370 501 – 512 800	77 362 + 31% of taxable income above 370 500
Primary rebate (under 65): R17 235 per year	

(7)

[35]

QUESTION 2: MEASUREMENT

2.1 An L-shaped floor is formed by removing a 2 m × 2 m square from one corner of a 6 m × 5 m rectangle. It will be tiled with 600 mm × 600 mm tiles. Allow 10% extra for breakages and cutting. Tiles are sold in boxes of 4 at R289 per box.

2.1.1 Calculate the floor area. (3)

2.1.2 Calculate the number of tiles needed, including the 10% extra. (4)

2.1.3 Calculate the number of boxes and the total cost. (3)

2.2 A rectangular swimming pool is 8 m long and 4 m wide, with an average depth of 1,5 m.

2.2.1 Calculate the volume of water in litres. (3)

2.2.2 A hosepipe delivers 25 ℓ per minute. How many hours will it take to fill the pool? (3)

2.2.3 Water costs R28,40 per kilolitre. Calculate the cost of filling the pool. (4)

[20]

QUESTION 3: MAPS AND ROUTE PLANNING

3.1 A family must drive from Durban to Kokstad and can choose between two routes. Their car uses 8 ℓ of petrol per 100 km, and petrol costs R23,45 per litre.

Route	Distance	Average speed	Toll fees
A: national road	312 km	100 km/h	none
B: mountain pass	268 km	75 km/h	R62

3.1.1 Calculate the travelling time for each route in hours and minutes. (4)

3.1.2 Calculate the petrol cost for each route. (6)

3.1.3 Calculate the total travel cost of route B, including the toll. (2)

3.1.4 Which route would you recommend? Give TWO reasons. (3)

[15]

QUESTION 4: DATA HANDLING

4.1 The table summarises the test results of two NC(V) Level 4 classes.

Statistic	Class A	Class B
Number of students	32	28
Mean	58%	62%

Median	61%	55%
Range	45%	70%
Pass rate (50% or more)	78%	64%

4.1.1 The lecturer of Class B says, "My class did better because our mean is higher." Use the data to argue against this statement. (4)

4.1.2 Which class performed more consistently? Explain. (2)

4.1.3 How many students in Class A passed? (2)

4.1.4 Why might the mean of Class B be higher than its median? (2)

4.2 A college enrolled 1 250 students in 2023 and 1 610 students in 2026.

4.2.1 Calculate the percentage increase in enrolment. (3)

4.2.2 Which type of graph would best show the enrolment for each year from 2020 to 2026? Give a reason. (2)

4.3 Give THREE ways in which statistics in an advertisement or a news report can be misleading. (3)

4.4 Explain why a survey of 20 people is less reliable than a survey of 2 000 people. (2)

[20]

QUESTION 5: PROBABILITY AND RISK

5.1 A raffle sells 500 tickets at R20 each. There is one prize worth R5 000.

5.1.1 What is the probability that a person who buys one ticket wins? (1)

5.1.2 Calculate the expected value of buying one ticket, and explain what it means. (3)

5.2 A bank card has a 4-digit PIN (digits 0 to 9, which may repeat). What is the probability that a thief guesses the PIN correctly on the first attempt? (2)

5.3 Statistics show that 1 in 25 cellphones in a city is stolen each year. Insurance for a R6 000 phone costs R89 per month. Use calculations to advise a student whether the insurance is worth it. (4)

[10]

TOTAL: 100

END OF QUESTION PAPER